ANDREA FUSIELLO

Curriculum vitae et studiorum

PERSONAL DATA:

- Born in Palmanova (Italy), on May 17, 1968
- Nationality: Italian; marital status: married.
- Home address: Udine (I).

EDUCATION:

- *Dottorato* (PhD) in Information Engineering, University of Trieste (I), February 12, 1999. Title: *"Three-dimensional Vision for Structure and Motion Estimation"*. Supervisor: Prof. V. Roberto. External supervisor Dr. E. Trucco (Heriot-Watt University, UK). Programme coordinator: Prof. G. Longo (University of Trieste).
- *Laurea* (Master) degree in Computer Science, University of Udine (I), March 2, 1994. Mark: 110/110. Dissertation title: *"Autonomous navigation and indoor map synthesis with uncertain sensory data"*. Supervisor: Prof. Vito Roberto. Co-supervisor: Dr. Bruno Caprile (IRST, Trento).

RESEARCH EXPERIENCE:

Appointments

- **Full Professor** (*Professore ordinario*). Department of Engineering and Architecture, University of Udine. Scientific disciplinary sector: ING-INF/05. October 2023 - present.
- **Associate Professor** (*Professore associato*). Department of Electrical, Management and Mechanical Engineering, University of Udine. Scientific disciplinary sector: ING-INF/05 (INF/01 until October 31, 2013). December 2011 - September 2023.
- **Visiting Professor** at IUT Le Creusot – Université de Bourgogne (France), April 2-16, 2017.
- **Associate Professor** (*Professore associato*). Computer Science. Dipartimento di Informatica, University of Verona, Verona. January 2005 - December 2011.
- **Visiting professor** at IUT Le Creusot - Université de Bourgogne (France), 1-30/10/2008.
- **Assistant Professor** (*Ricercatore*). Computer Science. Dipartimento di Informatica, University of Verona, Verona. October 2001 - December 2004.
- **Research Associate**. Computer Vision techniques applied to Visualization and Human Computer Interaction. Prof. Vittorio Murino. Dipartimento Scientifico e Tecnologico, University of Verona, Verona, IT. September 1999 - September 2001.
- **EPSRC Visiting Research Fellow**. Video mosaicing and uncalibrated reconstruction. Dr. Emanuele Trucco. Department of Computing and Electrical Engineering, Heriot-Watt University, Edinburgh, UK. April 1999 - September 1999.
- **Marie Curie Fellowship**, European Commission, for a six-month stay at Heriot-Watt University, Edinburgh. Declined in favour of the EPSRC fellowship.
- **Research Fellow**. Model-based vision (pose estimation, registration) and integration of optical and acoustic data, within the BRITE-EURAM project "VENICE" (Virtual Environment Interface by Sensory Integration for Inspection and Manipulation Control in Multifunctional Underwater Vehicles). Prof. V. Murino. University of Udine. November 1998 - March 1999.
- **Doctoral student**. Computer Vision: 3D reconstruction from stereo and uncalibrated motion. Prof. V. Roberto, Dr. E. Trucco. Dipartimento di Matematica e Informatica, University of Udine. November 1995 - October 1998.

- **Socrates Fellow**. Research related to the doctoral thesis, hosted by Dr. Emanuele Trucco, Department of Computing and Electrical Engineering, Heriot-Watt University, Edinburgh. September - December 1997.
- **Research contract**. Processor scheduling and allocation in distributed systems and fault-tolerance schemes, supervised by Prof. Alan Bertossi, Department of Mathematics, University of Trento. September 1995 - February 1996.

PROJECTS:

- **Principal investigator**
  - *POC-MISE: Proof of Concept funding for patent valorisation.* Principal investigator. Funding: EUR 33,000. 2021-2022.
  - *Automatic profilometry system for monitoring transportation in port areas.* POR FESR 2014-2020 (SPATA). Research unit leader. Funding: EUR 45,421. 2017-2018.
  - *Leveraging on Intensity images for Reverse Engineering.* "Joint Project" program of the University of Verona, 2009 (LIRE). Scientific coordinator. Funding: EUR 65,000.
  - *Three-Dimensional Shape Indexing and Retrieval Techniques.* Italian Ministry of Education, University and Research, PRIN 2006 (3-SHIRT). National scientific coordinator. Funding: EUR 73,000.
  - *CooperInt 2004.* A program of the University of Verona, for developing international liaisons. €9,600.
  - *Retrieval, Integration and Analysis of Hypermedia Data* (RIADI). University of Verona, Young Researchers Programme, 2000. Hypermedia information retrieval through data analysis, query languages and abstract interpretation.
  - *Industry funded projects* for a total amount of approximately EUR 200,000.
- **Co-investigator**
  - *Suspicious and Abnormal behavior Monitoring Using a netwoRk of cAmeras for sItuation awareness*, FP7-SEC-2007-01 Grant No. 217899, European Union (SAMURAI), 2008-2010.
  - *Low cost automatic system for 3D imaging and modelling*, Ministry of Education, University and Research, COFIN 2004 (LIMA3D), 2004-2005.
  - *Augmented Reality for Remotely Operated Vehicles based on 3D acoustical and optical sensors for underwater inspection and survey*, GROWTH project no. GRD1-2000-25409, European Union (ARROV), 2001-2003. Proposal co-author and participant.
  - *Augmented Reality for Teleoperation of Free Flying Robots*, ASI (Italian Space Agency), Scientific Research Grant no. 391, 2000 (AUREA), 2001-2003. Initiator, proposal co-author and participant.
  - *The Sounding Landscape* (SoL), Hewlett Packard Philanthropic Programme, 2001-2002. Visual analysis and sound synthesis to enable blind users to hear a scene. Proposal co-author and participant.
  - *Virtual Environment Interface by Sensory Integration for Inspection and Manipulation Control in Multifunctional Underwater Vehicles* (VENICE), BRPR-CT96-0150, European Commission, BRITE-EURAM III, 1996-1999. Integration of acoustic and optical data for underwater teleoperation. Participant.
  - *Representation and Processing of Spatial Data in Geographic Information Systems*, Ministry of University and Scientific and Technological Research, COFIN 2000 (SPADA), 2001-2002. Proposal co-author.

TECHNOLOGY TRANSFER:

- Doctoral scholarship (36th cycle) funded by SeeThrough s.r.l., Brusaporto (Bergamo).

- One-year research fellowship funded by Comark s.r.l., Udine, 2021.
- One-year research fellowship funded by Helica s.r.l., Amaro (Udine), 2019.
- One-year research fellowship in collaboration with 3Dflow s.r.l., funded by the European Social Fund project "Higher education and development" (HEaD), 2017.
- Doctoral scholarship (28th cycle) funded by ST Microelectronics, Agrate (Monza and Brianza).
- Co-founder of 3Dflow s.r.l., a spin-off of the University of Verona (until 2011) and the University of Udine (until 2016).

PUBLICATIONS:

Journal papers

1. Federica Arrigoni, Kathlén Kohn, Andrea Fusiello, and Tomas Pajdla. *An Algebraic Geometry Approach to Viewing Graph Solvability.* **IEEE Transactions on Pattern Analysis and Machine Intelligence** 48(7) (2026), 7229–7240.
2. Sofia Baldini, Filippo Stazi, Riccardo Bernardini, Andrea Fusiello, Paolo Gardonio, and Roberto Rinaldo. *Vibration measurement with neuromorphic vision sensors.* **Mechanical Systems and Signal Processing** 249 (2026), 114102.
3. Sadiq Olayiwola Macaulay, Eleonora Maset, Francesco Boscutti, Paolo Cingano, Francesco Trevisan, Giacomo Trotta, Marco Vuerich, and Andrea Fusiello. *Transferring RGB-Pretrained CNNs to Multispectral UAV Imagery for Salt Marsh Vegetation Classification.* **Remote Sensing** 18(4) (2026).
4. Federica Arrigoni, Andrea Fusiello, Romeo Rizzi, Elisa Ricci, and Tomas Pajdla. *Revisiting Viewing Graph Solvability: an Effective Approach Based on Cycle Consistency.* **IEEE Transactions on Pattern Analysis and Machine Intelligence** 47(5) (2025), 3271–3284.
5. Andrea Fusiello and Pier Luca Montessoro. *Distributed Interval Synchronization.* **IEEE Access** 12 (2024), 1700–1705.
6. Sadiq Olayiwola Macaulay, Eleonora Maset, and Andrea Fusiello. *Classification of compressed full-waveform airborne lidar data.* **Remote Sensing Letters** 15(7) (2024), 729–738.
7. Eleonora Maset and Andrea Fusiello. *Scales of oblique photographs updated.* **ISPRS Journal of Photogrammetry and Remote Sensing** 207 (2024), 198–202.
8. Eleonora Maset, Luca Magri, and Andrea Fusiello. *Principled bundle block adjustment with multi-head cameras.* **ISPRS Open Journal of Photogrammetry and Remote Sensing** 11 (2024), 100051.
9. P. Gardonio, G. Guernieri, E. Turco, L. Dal Bo, R. Rinaldo, and A. Fusiello. *Reconstruction of the sound radiation field from flexural vibration measurements with multiple cameras.* **Mechanical Systems and Signal Processing** 195 (2023), 110289.
10. Paolo Gardonio, Roberto Rinaldo, Loris Dal Bo, Roberto Del Sal, Emanuele Turco, and Andrea Fusiello. *Free-field sound radiation measurement with multiple synchronous cameras.* **Measurement** 188 (2022), 110605.
11. Roberto Del Sal, Loris Dal Bo, Emanuele Turco, Andrea Fusiello, Alessandro Zanarini, Roberto Rinaldo, and Paolo Gardonio. *Structural vibration measurement with multiple synchronous cameras.* **Mechanical Systems and Signal Processing** 157 (2021), 107742.
12. Federica Arrigoni and Andrea Fusiello. *Synchronization Problems in Computer Vision with Closed-Form Solutions.* **International Journal of Computer Vision** 128(1) (January 2020), 26–52.
13. E. Maset, L. Scalera, D. Zonta, I.M. Alba, F. Crosilla, and A. Fusiello. *Procrustes analysis for the virtual trial assembly of large-size elements.* **Robotics and Computer-Integrated Manufacturing** 62 (2020), 101885.
14. F. Arrigoni and A. Fusiello. *Bearing-based Network Localizability: A Unifying View.* **IEEE Transactions on Pattern Analysis and Machine Intelligence** 41(9) (2019), 2049–2069.
15. Antonio Canclini, Francesco Malapelle, Marco Marcon, Stefano Tubaro, and Andrea Fusiello. *View-synthesis from uncalibrated cameras and parallel planes.* **Signal Processing: Image Communication** 79 (2019), 40–53.
16. S. Zorzi, E. Maset, A. Fusiello, and F. Crosilla. *Full-Waveform Airborne LiDAR Data Classification Using Convolutional Neural Networks.* **IEEE Transactions on Geoscience and Remote Sensing** 57(10) (2019), 8255–8261.

17. Federica Arrigoni, Beatrice Rossi, Pasqualina Fragneto, and Andrea Fusiello. *Robust Synchronization in SO(3) and SE(3) via Low-rank and Sparse Matrix Decomposition.* **Computer Vision and Image Understanding** 174 (2018), 95–113.
18. Luca Magri and Andrea Fusiello. *Multiple structure recovery with maximum coverage.* **Machine Vision and Applications** 29(1) (2018), 159–173.
19. Tommaso Banelli, Marco Vuano, Federico Fogolari, Andrea Fusiello, Gennaro Esposito, and Alessandra Corazza. *Automation of peak-tracking analysis of stepwise perturbed NMR spectra.* **Journal of Biomolecular NMR** (2017), 1–14.
20. Luca Magri and Andrea Fusiello. *Multiple structure recovery via robust preference analysis.* **Image and Vision Computing** 67 (2017), 1–15.
21. Luca Magri and Andrea Fusiello. *Multiple structure recovery with T-linkage.* **Journal of Visual Communication and Image Representation** 49 (2017), 57–77.
22. E. Maset, F. Crosilla, and A. Fusiello. *Errors-in-Variables Anisotropic Extended Orthogonal Procrustes Analysis.* **IEEE Geoscience and Remote Sensing Letters** 14(1) (2017), 57–61.
23. Federica Arrigoni, Beatrice Rossi, and Andrea Fusiello. *Spectral Synchronization of Multiple Views in SE(3).* **SIAM Journal on Imaging Sciences** 9(4) (2016), 1963–1990.
24. Francesco Malapelle, Andrea Fusiello, Beatrice Rossi, and Pasqualina Fragneto. *A data-fusion approach to motion-stereo.* **Signal Processing: Image Communication** 43 (April 2016), 42–53.
25. Andrea Fusiello and Fabio Crosilla. *Solving Bundle Block Adjustment by Generalized Anisotropic Procrustes Analysis.* **ISPRS Journal of Photogrammetry and Remote Sensing** 102 (April 2015), 209–221.
26. Roberto Toldo, Riccardo Gherardi, Michela Farenzena, and Andrea Fusiello. *Hierarchical structure-and-motion recovery from uncalibrated images.* **Computer Vision and Image Understanding** 140 (November 2015), 127–143.
27. M. Corsini, M. Dellepiane, F. Ganovelli, R. Gherardi, A. Fusiello, and R. Scopigno. *Fully Automatic Registration of Image Sets on Approximate Geometry.* **International Journal of Computer Vision** 102(1-3) (March 2013), 91–111.
28. Said Pertuz, Domenec Puig, Miguel Angel Garcia, and Andrea Fusiello. *Generation of All-in-Focus Images by Noise-Robust Selective Fusion of Limited Depth-of-Field Images.* **IEEE Transactions on Image Processing** 22(3) (March 2013), 1242–1251.
29. Roberto Toldo and Andrea Fusiello. *Image-consistent patches from unstructured points with J-linkage.* **Image and Vision Computing** 31(10) (2013), 756–770.
30. A. Fusiello and L. Irsara. *Quasi-Euclidean epipolar rectification of uncalibrated images.* **Machine Vision and Applications** 22(4) (2011), 663–670.
31. A. Colombari and A. Fusiello. *Patch-Based Background Initialization in Heavily Cluttered Video.* **IEEE Transactions on Image Processing** 19(4) (April 2010), 926–933.
32. R. Toldo, U. Castelllani, and A. Fusiello. *The bag of words approach for retrieval and categorization of 3D objects.* **The Visual Computer** 26(10) (October 2010), 1257–1268.
33. Michela Farenzena and Andrea Fusiello. *Stabilizing 3D modeling with geometric constraints propagation.* **Computer Vision and Image Understanding** 113(11) (2009), 1147–1157.
34. U. Castellani, R. Gherardi, A. Fusiello, and V. Murino. *Automatic selection of MRF control parameters by Reactive Tabu Search.* **Image and Vision Computing** 25(11) (2007), 1824–1832.
35. A. Colombari, A. Fusiello, and V. Murino. *Segmentation and Tracking of Multiple Video Objects.* **Pattern Reconition** 40(4) (2007), 1307–1317.
36. A. Fusiello. *A matter of notation: Several uses of the Kronecker product in 3-D computer vision.* **Pattern Recognition Letters** 28(15) (2007), 2127–2132.
37. A. Fusiello. *Specifying Virtual Cameras in Uncalibrated View Synthesis.* **IEEE Transactions on Circuits and Systems for Video Technology** 17(5) (2007), 604–611.
38. U. Castellani, A. Fusiello, V. Murino, L. Papaleo, E. Puppo, and M. Pittore. *A complete system for on-line 3D modelling from acoustic images.* **Signal Processing - Image Communication** 20(9-10) (2005), 832–852.
39. A. Fusiello, M. Farenzena, A. Busti, and A. Benedetti. *Computing Rigorous Bounds to the Accuracy of Calibrated Stereo Reconstruction.* **IEE Proceedings. Vision, Image and Signal Processing** 152(6) (December 2005), 695–701.

40. A. Fusiello, A. Benedetti, M. Farenzena, and A. Busti. *Globally Convergent Autocalibration using Interval Analysis*. **IEEE Transactions on Pattern Analysis and Machine Intelligence** 26(12) (December 2004), 1633–1638.
41. A. Fusiello and V. Murino. *Augmented Scene Modeling and Visualization by Optical and Acoustic Sensor Integration*. **IEEE Transactions on Visualization and Computer Graphics** 10(6) (December 2004), 625–636.
42. U. Castellani, A. Fusiello, and V. Murino. *Registration of Multiple Acoustic Range Views for Underwater Scene Reconstruction*. **Computer Vision and Image Understanding** 87(3) (2002), 78–89.
43. F. Odone, A. Fusiello, and E. Trucco. *Layered Representation of a Video Shot with Mosaicing*. **Pattern Analysis and Applications** 5(3) (2002), 296–305.
44. A. Fusiello. *Uncalibrated Euclidean Reconstruction: A Review*. **Image and Vision Computing** 18(6-7) (May 2000), 555–563.
45. A. Fusiello, V. Roberto, and E. Trucco. *Symmetric Stereo with Multiple Windowing*. **International Journal of Pattern Recognition and Artificial Intelligence** 14(8) (2000), 1053–1066.
46. A. Fusiello, E. Trucco, and A. Verri. *A Compact Algorithm for Rectification of Stereo Pairs*. **Machine Vision and Applications** 12(1) (2000), 16–22.
47. A. Fusiello, E. Trucco, T. Tommasini, and V. Roberto. *Improving Feature Tracking with Robust Statistics*. **Pattern Analysis and Applications** 2(4) (1999), 312–320.
48. E. Trucco, A. Fusiello, and V. Roberto. *Robust Motion and Correspondence of Noisy 3-D Point Sets with Missing Data*. **Pattern Recognition Letters** 20(9) (September 1999), 889–898.
49. A. A. Bertossi and A. Fusiello. *Rate-Monotonic scheduling for hard-real-time systems*. **European Journal of Operational Research** 96(3) (February 1997), 429–443.
50. A. Fusiello and B. Caprile. *Synthesis of Indoor Maps in Presence of Uncertainty*. **Robotics and Autonomous Systems** 22(2) (November 1997), 103–114.

Books

51. Andrea Fusiello. *Computer Vision: Three-dimensional Reconstruction Techniques*. Springer, 2024. ISBN: 978-3-031-34506-7. DOI: 10.1007/978-3-031-34507-4.
52. Fabio Crosilla, Alberto Beinat, Andrea Fusiello, Eleonora Maset, and Domenico Visintini. *Advanced Procrustes Analysis Models in Photogrammetric Computer Vision*. Vol. 590. CISM International Centre for Mechanical Sciences. Springer, Cham, 2019. ISBN: 978-3-030-11759-7. DOI: 10.1007/978-3-030-11760-3.
53. Andrea Fusiello. *Visione computazionale. Tecniche di ricostruzione tridimensionale*. II. Franco Angeli, Milano, 2018. ISBN: 9788891771933.

Editorships

54. *Guest Editorial: Special Issue on Traditional Computer Vision in the Age of Deep Learning*. Vol. 132. 9. September 2024, pp. 4128–4130. DOI: 10.1007/s11263-024-02062-2.
55. Gian Luca Foresti, Andrea Fusiello, and Edwin R. Hancock, eds. *Image Analysis and Processing - ICIAP 2023 - Proceedings, Part I-II*. Vol. 14233. Lecture Notes in Computer Science. Springer, 2023. ISBN: 978-3-031-43147-0. DOI: 10.1007/978-3-031-43148-7.
56. Adrien Bartoli and Andrea Fusiello, eds. *Computer Vision - ECCV 2020 Workshops Proceedings, Part I-VI*. Vol. 12535-12540. Lecture Notes in Computer Science. Springer, 2020. ISBN: 978-3-030-66414-5. DOI: 10.1007/978-3-030-66415-2.
57. Andrea Fusiello and Oliver Bimber, eds. *Eurographics 2019 - Posters*. The Eurographics Association, 2019. DOI: 10.2312/egp.20192004.
58. Andrea Fusiello, Vittorio Murino, and Rita Cucchiara, eds. *Computer Vision - ECCV 2012. Workshops and Demonstrations Proceedings, Part I-III*. Vol. 7583-7585. Lecture Notes in Computer Science. Springer, 2012. ISBN: 978-3-642-33862-5. DOI: 10.1007/978-3-642-33863-2.
59. Andrea Fusiello and Michael Wimmer, eds. *Eugraphics 2012 - Posters*. Eurographics Association, 2012.
60. Hongbin Zha, H. Saito, V. Murino, and A. Fusiello, eds. *IEEE Transactions on Systems, Man and Cybernetics, Part B*. Vol. 33. 4. Special issue on 3-D image analysis and modeling. August 2003, pp. 550–553. DOI: 10.1109/TSMCB.2003.814779.

Book chapters

61. Fabio Crosilla, Eleonora Maset, and Andrea Fusiello. "Procrustean Photogrammetry: From Exterior Orientation to Bundle Adjustment". In: *New Advanced GNSS and 3D Spatial Techniques: Applications to Civil and Environmental Engineering, Geophysics, Architecture, Archeology and Cultural Heritage*. Ed. by R. Cefalo, J.B. Zieliński, and M. Barbarella. Lecture Notes in Geoinformation and Cartography. Springer International Publishing, 2018, pp.157–165. ISBN: 978-3-319-56218-6. DOI: 10.1007/978-3-319-56218-6_12.
62. Andrea Fusiello. "Lemmi vari". In: *Dizionario di Informatica ICT e media digitali*. Ed. by Edoardo Fleischner. La Piccola Treccani, Scienza e Tecnica. Treccani, Roma, 2012.
63. S. Ivekovic, A. Fusiello, and E. Trucco. "Fundamentals of Multiple-View Geometry". In: *3D Videocommunication. Algorithms, concepts and real-time systems in human centered communication*. Ed. by O. Schreer, P. Kauff, and T. Sikora. John Wiley & Sons, 2005. Chap. 6. ISBN: 9780470022719. DOI: 10.1002/0470022736.ch6.
64. A. Trucco, M. Palmese, A. Fusiello, and V. Murino. "Three-dimensional underwater acoustical imaging and processing". In: *Underwater Acoustic Digital Signal Processing and Communication Systems*. Ed. by R.S.H. Istepanian and M. Stojanovic. Kluwer Academic, 2002, pp.247–274. ISBN: 978-0-7923-7304-9.

Selected conferences

65. Federica Arrigoni, Andrea Fusiello, and Tomas Pajdla. *A Direct Approach to Viewing Graph Solvability*. In: **Proceedings of the 18th European Conference on Computer Vision (ECCV)**. (Oral, acceptance rate: 2.3%). Springer Nature Switzerland, 2024, pp.319–334. ISBN: 978-3-031-73232-4. DOI: 10.1007/978-3-031-73232-4_18.
66. Rakshith Madhavan, Andrea Fusiello, and Federica Arrigoni. *Synchronization of Projective Transformations*. In: **Proceedings of the 18th European Conference on Computer Vision (ECCV)**. Acceptance rate: 27.9%). Springer Nature Switzerland, 2024, pp.18–36. ISBN: 978-3-031-72913-3. DOI: 10.1007/978-3-031-72913-3_2.
67. Federica Arrigoni, Tomas Pajdla, and Andrea Fusiello. *Viewing Graph Solvability in Practice*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)**. (Oral, acceptance rate: 2.4%. Shortlisted for best paper 17/2161). October 2023, pp.8147–8155. DOI: 10.1109/ICCV51070.2023.00748.
68. GK Tejus, Giacomo Zara, Paolo Rota, Andrea Fusiello, Elisa Ricci, and Federica Arrigoni. *Rotation Synchronization via Deep Matrix Factorization*. In: **2023 IEEE International Conference on Robotics and Automation (ICRA)**. 2023, pp.2113–2119. DOI: 10.1109/ICRA48891.2023.10160548.
69. Andrea Fusiello. *Exact Affine Histogram Matching by Cumulants Transformation*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (May 23–27, 2022). Lecture Notes in Computer Science. (Acceptance rate: 54%.) Springer-Verlag, 2022, pp.199–210. ISBN: 978-3-031-06429-6. DOI: 10.1007/978-3-031-06430-2_17.
70. Federica Arrigoni, Andrea Fusiello, Elisa Ricci, and Tomas Pajdla. *Viewing Graph Solvability via Cycle Consistency*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)** (October 11–17, 2021). (Oral, acceptance rate: 3.4%. Honorable mention award). 2021, pp.5540–5549. DOI: 10.1109/ICCV48922.2021.00549.
71. Andrea Porfiri Dal Cin, Luca Magri, Federica Arrigoni, Andrea Fusiello, and Giacomo Boracchi. *Synchronization of Group-Labelled Multi-Graphs*. In: **Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)** (October 11–17, 2021). (Acceptance rate: 26%.) 2021, pp.6453–6463. DOI: 10.1109/ICCV48922.2021.00639.
72. Marco Patané and Andrea Fusiello. *Vehicle Classification from Profile Measures*. In: **Proceedings of the 25th International Conference on Pattern Recognition (ICPR)** (January 10–15, 2021). (Oral, acceptance rate: 4.4%). IEEE Computer Society Press, 2021, pp.6656–6663. DOI: 10.1109/ICPR48806.2021.9413185.
73. E. Maset, L. Magri, I. Toschi, and A. Fusiello. *Bundle Block Adjustment With Constrained Relative Orientations*. In: **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences** (August 31–September 2, 2020). Vol. V-2-2020. (Acceptance rate: 19%. Best student paper award). 2020, pp.49–55. DOI: 10.5194/isprs-annals-V-2-2020-49-2020.
74. Luca Magri and Andrea Fusiello. *Fitting Multiple Heterogeneous Models by Multi-Class Cascaded T-Linkage*. In: **2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)** (June 16–20, 2019). (Acceptance rate: 25.1%). 2019, pp.7452–7460. DOI: 10.1109/CVPR.2019.00764.

75. L. Magri and A. Fusiello. *Reconstruction of interior walls from point cloud data with min-hashed J-linkage.* In: **Proceedings of the International Conference on 3D Vision (3DV)** (September 5–8, 2018). (Acceptance rate: 48.8%). 2018, pp.131–139. DOI: 10.1109/3DV.2018.00025.
76. Federica Arrigoni, Eleonora Maset, and Andrea Fusiello. *Synchronization in the Symmetric Inverse Semigroup.* In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 11–15, 2017). Lecture Notes in Computer Science. (Acceptance rate: 60%). Springer International Publishing, 2017, pp.70–81. DOI: 10.1007/978-3-319-68548-9_7.
77. Eleonora Maset, Federica Arrigoni, and Andrea Fusiello. *Practical and Efficient Multi-View Matching.* In: **The IEEE International Conference on Computer Vision (ICCV)** (October 22, 2017). (Spotlight acceptance rate: 4.7%). 2017, pp.4568–4576. DOI: 10.1109/ICCV.2017.489.
78. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *Camera Motion from Group Synchronization.* In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 25–28, 2016). (Acceptance rate: 44.4%). IEEE, 2016, pp.546–555. DOI: 10.1109/3DV.2016.64.
79. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *Global Registration of 3D Point Sets via LRS decomposition.* In: **Proceedings of the 14th European Conference on Computer Vision (ECCV)** (October 8–16, 2016). (Acceptance rate: 26.6%). 2016, pp.489–504. DOI: 10.1007/978-3-319-46493-0_30.
80. A. Fusiello and F. Crosilla. *Fast and Resistant Procrustean Bundle Adjustment.* In: **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences** (July 12–19, 2016). Vol. III. (Acceptance rate: 54%). 2016, pp.35–41. DOI: doi:10.5194/isprs-annals-III-3-35-2016.
81. Luca Magri and Andrea Fusiello. *Multiple Models Fitting as a Set Coverage Problem.* In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 27–July 1, 2016). (Acceptance rate: 30%). 2016, pp.3318–3326. DOI: 10.1109/CVPR.2016.361.
82. Federica Arrigoni, Andrea Fusiello, and Beatrice Rossi. *On Computing the Translations Norm in the Epipolar Graph.* In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 19–22, 2015). (Acceptance rate: 46%). IEEE, 2015, pp.300–308. DOI: 10.1109/3DV.2015.41.
83. Andrea Fusiello, Fabio Crosilla, and Francesco Malapelle. *Procrustean point-line registration and the NPnP problem.* In: **Proceedings of the International Conference on 3D Vision (3DV)** (October 19–22, 2015). (Oral, acceptance rate: 15%). IEEE, 2015, pp.250–255. DOI: 10.1109/3DV.2015.35.
84. Luca Magri and Andrea Fusiello. *Robust Multiple Model Fitting with Preference Analysis and Low-rank Approximation.* In: **Proceedings of the British Machine Vision Conference (BMVC)** (September 7–10, 2015). Ed. by Xianghua Xie, Mark W. Jones, and Gary K. L. Tam. (Acceptance rate: 33%). BMVA Press, 2015, pp.20.1–20.12. ISBN: 1-901725-53-7. DOI: 10.5244/C.29.20.
85. F. Arrigoni, L. Magri, B. Rossi, P. Fragneto, and A. Fusiello. *Robust Absolute Rotation Estimation via Low-rank and Sparse Matrix Decomposition.* In: **Proceedings of the International Conference on 3D Vision (3DV)** (December 8–11, 2014). (Oral, acceptance rate: 15.5%). 2014, pp.491–498. DOI: 10.1109/3DV.2014.48.
86. Luca Magri and Andrea Fusiello. *T-Linkage: A Continuous Relaxation of J-Linkage for Multi-Model Fitting.* In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 24–27, 2014). (Acceptance rate: 29.88%). 2014, pp.3954–3961. DOI: 10.1109/CVPR.2014.505.
87. S. Fantoni, U. Castellani, and A. Fusiello. *Accurate and automatic alignment of range surfaces.* In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Oral, acceptance rate: 18%). 2012, pp.73–80. DOI: 10.1109/3DIMPVT.2012.63.
88. P. Fragneto, A. Fusiello, L. Magri, B. Rossi, and M. Ruffini. *Uncalibrated View Synthesis with Homography Interpolation.* In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Acceptance rate: 70%). 2012, pp.270–277. DOI: 10.1109/3DIMPVT.2012.39.
89. V. Garro, F. Crosilla, and A. Fusiello. *Solving the PnP Problem with Anisotropic Orthogonal Procrustes Analysis.* In: **Proceedings of the 2nd Joint 3DIM/3DPVT Conference: 3D Imaging, Modeling, Processing, Visualization and Transmission (3DIMPVT)** (October 13–15, 2012). (Acceptance rate: 70%). 2012, pp.262–269. DOI: 10.1109/3DIMPVT.2012.40.
90. Riccardo Gherardi, Michela Farenzena, and Andrea Fusiello. *Improving the Efficiency of Hierarchical Structure-and-Motion.* In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 13–18, 2010). (Acceptance rate: 27%). 2010, pp.1594–1600. DOI: 10.1109/CVPR.2010.5539782.

91. Riccardo Gherardi and Andrea Fusiello. *Practical Autocalibration*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (September 5–11, 2010). Lecture Notes in Computer Science. (Acceptance rate: 28%). Springer, 2010, pp.790–801. DOI: 10.1007/978-3-642-15549-9_57.
92. Roberto Toldo and Andrea Fusiello. *Photo-consistent Planar Patches from Unstructured Cloud of Points*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (September 5–11, 2010). Lecture Notes in Computer Science. (Acceptance rate: 28%). Springer, 2010, pp.589–602. DOI: 10.1007/978-3-642-15555-0_43.
93. M. Farenzena, A. Fusiello, and R. Gherardi. *Structure-and-motion Pipeline on a Hierarchical Cluster Tree*. In: **Proceedings of the IEEE International Workshop on 3-D Digital Imaging and Modeling (3DIM)** (October 3–4, 2009). 2009, pp.1489–1496. ISBN: 978-1-4244-4441-0. DOI: 10.1109/ICCVW.2009.5457435.
94. Roberto Toldo and Andrea Fusiello. *Automatic Estimation of the Inlier Threshold in Robust Multiple Structures Fitting*. In: **Proceedings of the 15th International Conference on Image Analysis and Processing (ICIAP)** (September 8–11, 2009). Vol. 5716. Lecture Notes in Computer Science. (Oral, acceptance rate: 21%. Best student paper award). Springer, 2009, pp.123–131. ISBN: 978-3-642-04145-7. DOI: 10.1007/978-3-642-04146-4_15.
95. M. Farenzena and A. Fusiello. *3D Surface Models by Geometric Constraints Propagation*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 24–26, 2008). (Acceptance rate: 32%). IEEE Computer Society, 2008, pp.1–8. DOI: 10.1109/CVPR.2008.4587776.
96. A. Fusiello and L. Irsara. *Quasi-Euclidean Uncalibrated Epipolar Rectification*. In: **Proceedings of the 19th International Conference on Pattern Recognition (ICPR)** (December 8–11, 2008). (Acceptance rate: 62%). 2008, pp.1–4. DOI: 10.1109/ICPR.2008.4761561.
97. R. Toldo and A. Fusiello. *Robust Multiple Structures Estimation with J-Linkage*. In: **Proceedings of the European Conference on Computer Vision (ECCV)** (October 12–18, 2008). Vol. 5302. Lecture Notes in Computer Science. (Acceptance rate: 29%). Springer, 2008, pp.537–547. DOI: 10.1007/978-3-540-88682-2_41.
98. A. Colombari, A. Fusiello, and V. Murino. *Video Objects Segmentation by Robust Background Modeling*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 10–14, 2007). (Oral, acceptance rate: 20%). IEEE Computer Society, 2007, pp.155–164. DOI: 10.1109/ICIAP.2007.4362773.
99. M. Farenzena and A. Fusiello. *Recovering Intrinsic Images using an Illumination Invariant Image*. In: **Proceedings of the IEEE International Conference on Image Processing (ICIP)** (September 16–19, 2007). Vol. 3. (Acceptance rate: 49%). 2007, pp.485–488. DOI: 10.1109/ICIP.2007.4379352.
100. A. Fusiello and L. Irsara. *An Uncalibrated View-Synthesis Pipeline*. In: **Proceedings of the International Conference on Image Analysis and Processing (ICIAP)** (September 10–14, 2007). (Oral, acceptance rate: 20%). IEEE Computer Society, 2007, pp.609–614. DOI: 10.1109/ICIAP.2007.4362844.
101. M. Farenzena, A. Fusiello, and A. Dovier. *Reconstruction with Interval Constraints Propagation*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 17–22, 2006). (Acceptance rate: 28%). IEEE Computer Society, 2006, pp.1185–1190. DOI: 10.1109/CVPR.2006.246.
102. A. Colombari, A. Fusiello, and V. Murino. *Uncalibrated Interpolation of Rigid Displacements for View Synthesis*. In: **Proceedings of the IEEE International Conference on Image processing (ICIP)** (September 11–14, 2005). (Acceptance rate: 46%). 2005, pp.1049–1052. DOI: 10.1109/ICIP.2005.1529934.
103. M. Farenzena, A. Busti, A. Fusiello, and A. Benedetti. *Rigorous accuracy bounds for calibrated stereo reconstruction*. In: **Proceedings of the 17th International Conference on Patter Recognition (ICPR)** (August 23–26, 2004). Vol. 4. (Oral, acceptance rate: 18%). IEEE Computer Society, 2004, pp.288–292. DOI: 10.1109/ICPR.2004.1333760.
104. R. Marzotto, A. Fusiello, and V. Murino. *High Resolution Video Mosaicing with Global Alignment*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 27–July 2, 2004). Vol. I. (Acceptance rate: 30%). IEEE Computer Society, 2004, pp.692–698. DOI: 10.1109/CVPR.2004.1315099.
105. A. Sanson, U. Castellani, A. Fusiello, and V. Murino. *Fast Model Tracking with Multiple Cameras for Augmented Reality*. In: **Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST)** (November 10–12, 2004). (Acceptance rate: 39%). 2004, pp.170–173. DOI: 10.1145/1077534.1077569.

106. A. Benedetti, A. Busti, M. Farenzena, and A. Fusiello. *Globally Convergent Autocalibration*. In: **Proceedings of the 9th IEEE International Conference on Computer Vision (ICCV)** (October 14–17, 2003). Vol. II. (Acceptance rate: 21%). IEEE Computer Society, 2003, pp.1426–1432. DOI: 10.1109/ICCV.2003.1238657.

107. A. Fusiello, M. Aprile, R. Marzotto, and V. Murino. *Mosaic of a video shot with multiple moving objects*. In: **Proceedings of IEEE International Conference on Image Processing (ICIP)** (September 14–18, 2003). Vol. II. (Acceptance rate: 43%). 2003, pp.307–310. DOI: 10.1109/ICIP.2003.1246678.

108. A. Fusiello, S. Caldrer, S. Ceglie, N. Mattern, and V. Murino. *View Synthesis from Uncalibrated Images Using Parallax*. In: **Proceedings of the 12th International Conference on Image Analysis and Processing (ICIAP)** (September 17–19, 2003). (Oral, acceptance rate: 26%). IEEE Computer Society, 2003, pp.146–151. DOI: 10.1109/ICIAP.2003.1234041.

109. A. Fusiello, U. Castellani, L. Ronchetti, and V. Murino. *Model Acquisition by Registration of Multiple Acoustic Range Views*. In: **Proceedings of the 7th European Conference on Computer Vision (ECCV)**. Ed. by A. Heyden, G. Sparr, M. Nielsen, and P. Johansen. Vol. 2351. Lecture Notes in Computer Science. (Acceptance rate: 38%). Springer, 2002, pp.805–819. DOI: 10.1007/3-540-47967-8_54.

110. A. Fusiello, A. Panuccio, V. Murino, F. Fontana, and D. Rocchesso. *A Multimodal Electronic Travel Aid Device*. In: **Proceedings of the Fourth IEEE International Conference on Multimodal Interfaces (ICMI)** (October 14–16, 2002). (Acceptance rate: 43%). IEEE Computer Society, 2002, pp.39–44. DOI: 10.1109/ICMI.2002.1166966.

111. V. Murino, L. Ronchetti, U. Castellani, and A. Fusiello. *Reconstruction of Complex Environments by Robust Pre-aligned ICP*. In: **Proceedings of the International Conference on 3-D Digital Imaging and Modeling (3DIM)** (May 28–June 1, 2001). IEEE Computer Society Press, 2001, pp.187–194. DOI: 10.1109/IM.2001.924432.

112. A. Valinetti, A. Fusiello, and V. Murino. *Model tracking for Video-based Virtual Reality*. In: **Proceedings of the 11th International Conference on Image Analysis and Processing (ICIAP)** (September 26–28, 2001). (Oral, acceptance rate: 28%). IEEE Computer Society Press, 2001, pp.372–377. DOI: 10.1109/ICIAP.2001.957038.

113. V. Murino, A. Fusiello, N. Iuretigh, and E. Puppo. *3D Mosaicing for Environment Reconstruction*. In: **Proceedings of the International Conference on Pattern Recognition (ICPR)** (September 3–8, 2000). Vol. 3. (Acceptance rate: 68%). IEEE Computer Society Press, 2000, pp.362–366. DOI: 10.1109/ICPR.2000.903559.

114. A. Censi, A. Fusiello, and V. Roberto. *Image Stabilization by Features Tracking*. In: **Proceedings of the 10th International Conference on Image Analysis and Processing (ICIAP)** (September 27–29, 1999). (Acceptance rate: 57%). IEEE Computer Society Press, 1999, pp.665–667. DOI: 10.1109/ICIAP.1999.797671.

115. A. Fusiello, R. Giannitrapani, V. Isaia, and V. Murino. *Virtual Environment Modeling by Integrated Optical and Acoustic Sensing*. In: **Proceedings of the 2nd International Conference on 3-D Digital Imaging and Modeling (3DIM)** (October 4–8, 1999). IEEE Computer Society Press, 1999, pp.437–446. DOI: 10.1109/IM.1999.805375.

116. T. Tommasini, A. Fusiello, E. Trucco, and V. Roberto. *Making Good Features Track Better*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 23–25, 1998). (Acceptance rate: 30%). IEEE Computer Society Press, 1998, pp.178–183. DOI: 10.1109/CVPR.1998.698606.

117. A.A. Bertossi, A. Fusiello, and L.V. Mancini. *Fault-Tolerant Deadline-Monotonic Algorithm for Scheduling Hard-Real-Time Tasks*. In: **Proceedings of the 11th International Parallel Processing Symposium** (April 1–5, 1997). (Acceptance rate: 38%). IEEE Computer Society Press, 1997, pp.133–138. DOI: 10.1109/IPPS.1997.580868.

118. A. Fusiello, V. Roberto, and E. Trucco. *Efficient Stereo with Multiple Windowing*. In: **Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)** (June 17–19, 1997). (Acceptance rate: 32%). IEEE Computer Society Press, 1997, pp.858–863. DOI: 10.1109/CVPR.1997.609428.

119. A. Fusiello, E. Trucco, and A. Verri. *Rectification with Unconstrained Stereo Geometry*. In: **Proceedings of the British Machine Vision Conference (BMVC)** (September 8–11, 1997). Ed. by A. F. Clark. (Acceptance rate: 62%). BMVA Press, GB, 1997, pp.400–409.

The full list of papers with links to electronic copies is available at:
https://fusiello.github.io/index.php/Publications.html.

Citation indices from Google Scholar (July 2022):

Citation indices	All	Since 2017
Citations	7423	2402
h-index	40	24
i10-index	105	58

Updated bibliometric data, including VQR and ASN indicators, are available at: `https://fusiello.github.io/index.php/Bibliometrica.html`.

Patents

- Yash Singh, Roberto Toldo, Luca Magri, Simone Fantoni, and Andrea Fusiello. *Method for 3D modelling based on structure from motion processing of sparse 2D images*. U.S. Patent US 10,198,858 B2. February 2019.
- Fabio Crosilla, Andrea Fusiello, Eleonora Maset, and Stefano Zorzi. *Apparato e Metodo di Classificazione di Dati a Forma d'onda Completa da Segnali Retroriflessi*. IT102018000005375. 15 Maggio 2018.

Awards and distinctions

- Marr Prize Best Paper Honorable Mention, ICCV 2021.
- IEEE Senior Member since 2021.
- Regional Innovation Award promoted by Veneto Innovazione, 1999-2000.

INVITED SEMINARS:

Invited talks

Invited talks

- *Synchronisation: from pairwise measures to global values*. Apple Inc., Cupertino CA, June 16, 2019.
- *Synchronisation: from pairwise measures to global values*. Czech Institute of Informatics, Robotics and Cybernetics, Czech Technical University in Prague, July 16, 2019.
- *The long march of 3D reconstruction: from tabletop to outer space*. Johannes Kepler University Linz - Institute of Computer Graphics, November 20, 2018.
- *Synchronization problems*. Università di Modena e Reggio Emilia - Dipartimento di Ingegneria "Enzo Ferrari", October 21, 2016.
- *Synchronization problems*. European Centre for Living Technology (ECLT), Venice, October 14, 2016.
- *Synchronization problems*. IUT Le Creusot - Unviversité de Bourgogne, France, December 10, 2015.
- *Structure from motion: state of the art, examples and open issues*. Uppsala University (Sweden), May 27, 2014.
- *Structure and motion with Samantha*. Escola Tecnica Superior d'Enginyeria, Universitat Rovira i Virgili, Tarragona (Spain), November 11, 2011.
- *Structure and motion with Samantha*. Istituto Italiano di Tecnologia, Genova, October 17, 2011.
- *Recupero di struttura e di moto da immagini non calibrate*. IASF/INAF, Palermo, December 18, 2009.
- *Robust Multiple Structures Estimation and Architectural Modelling*. Dipartimento di Sistemi e Informatica dell'Università di Firenze, May 29, 2009.

- *Robust Multiple Structures Estimation with J-linkage*. LASMEA (CNRS) laboratory in Clermont-Ferrand (France), October 24, 2008.
- *Robust Multiple Structures Estimation with J-linkage*. IUT Le Creusot - Unviversité de Bourgogne (France), October 7, 2008.
- *Propagation of geometric constraints in 3D modelling*. CRS4, Pula (Cagliari), April 17, 2007.
- *3-D Reconstruction from Images*. University of California, Santa Barbara (USA), June 9, 2006.
- *Reconstruction with Interval Constraints Propagation*. University of California, Los Angeles (USA), June 12, 2006.
- *Collineations and image mosaics*. Heriot-Watt University, Edinburgh (UK), May 9, 2003.
- *Image-based Rendering*. Heriot-Watt University, Edinburgh (UK), May 15, 2003.
- *Il progetto VENICE: realtà aumentata per veicoli subacquei telecomandati*. Dipartimento di Matematica, Università di Bologna, April 5, 2000.
- *Prospettive di ricerca tra Visione e Grafica*. DISI, Università di Genova, November 24, 2000.
- *L'algoritmo SMW per il calcolo della disparità stereo*. DISI, Università di Genova, April 30, 1998.
- *Robust Feature Tracking*. California Institute of Technology, Pasadena (USA), July 2, 1998.
- *Rectification of Stereo Pairs*. Heriot-Watt University, Edinburgh (UK), November 17, 1997.
- *Un algoritmo per il calcolo della disparità stereo*. IRST, Trento, February 15, 1997.
- *A Symmetry Based Stereo Algorithm*. Heriot-Watt University, Edinburgh (UK), December 18, 1996.

TEACHING EXPERIENCE:

Undergraduate teaching

- Fundamentals of Computer Science (6 ECTS), Mechanical Engineering, University of Udine, since academic year 2014/2015.
- Fundamentals of Computer Science (6 ECTS), Civil Engineering, University of Udine, since academic year 2012/2013.
- Elements of Computer Science (6 ECTS), Architectural Sciences, University of Udine, academic years 2011/2012 and 2012/2013.
- Computer Graphics (6 ECTS), Computer Science, University of Verona, 2009/2010 and 2010/2011.
- Computer Graphics (5 ECTS), Multimedia Computer Science (formerly Information Technology), University of Verona, 2002/2003 - 2008/2009.
- Computational Geometry: Advanced Module (2 ECTS), Applied Mathematics, University of Verona, 2008/2009.
- Basic Computing Laboratory (2 ECTS), Computer Science, University of Verona, 2007/2008 and 2008/2009.
- Human-Computer Interaction (3 ECTS), Communication Science, University of Verona, 2005/2006 and 2006/2007.
- Programming Laboratory (4 ECTS), Computer Science, University of Verona, 2001/2002.
- Computing Laboratory accompanying the Programming course (60 hours), Computer Science, University of Verona, contract lecturer, 1998/1999 - 2000/2001.
- Integrated Course in Medical Physics, Statistics and Computer Science (15 hours), Nursing Diploma, University of Udine, Pordenone campus, contract lecturer, 1998/1999.

Graduate teaching

- Computer Vision (6 ECTS), Electronic Engineering, University of Udine, since 2012/2013 (the course title has changed over time).
- Digital Archaeology (6 ECTS), Ancient Civilisations, University of Udine, since 2016/2017, offered every other year.
- Computational Vision (6 ECTS), Computer Science and Engineering, University of Verona, 2009/2010 and 2010/2011.
- Computational Vision (5 ECTS), Intelligent and Multimedia Systems, University of Verona, 2002/2003 - 2008/2009.
- Computer Vision (5 ECTS), Computer Science degree programme, University of Verona, 2001/2002.
- Communication and Interaction (6 ECTS), Electronic and Multimedia Publishing, University of Verona, 2004/2005.

Scuola Superiore (University honours school)

- Advanced Mathematical Methods for Engineering Problems (14 hours), University of Udine, 2021/2022.
- Neural Networks (28 hours), University of Udine, 2013/2014.

Invited lectures

- *Introduction to scientific computing in MATLAB.* Scientific Skills Training Courses, PhD programme in Industrial and Information Engineering, University of Udine, 2013-2015.
- *Structure from motion in computer vision and photogrammetry.* Advanced Professional Training, International Centre for Mechanical Sciences (CISM), Udine, July 8, 2022.
- *Photogrammetric computer vision with cultural heritage applications.* VISMAC Summer School, Palermo, September 23, 2021.
- *Large-Scale Camera Estimation in 3D Vision: lab session.* VISMAC Summer School, Vico Equense, August 31, 2018.
- *Lab activity on 3D computer vision.* VISMAC Summer School, Grado, June 17, 2016.
- *Computer Graphics and 3D Reconstruction* (24 hours). Second-level postgraduate programme in Visual Computing and Multimedia Technology, University of Modena and Reggio Emilia, 2016/2017.
- *Elements of Photogrammetric Computer Vision.* VIBOT (Erasmus Mundus) Master's programme, IUT Le Creusot – Université de Bourgogne, April 2-16, 2017.
- *The devil is in the details: anatomy of a structure-from-motion pipeline.* Tutorial at the *International Computer Vision Summer School*, 14-20 July 2013, Calabria, IT.
- *Introduction to image-based modeling.* Tutorial at the VISMAC IAPR Summer School, 23 October 2012, Genova, IT.
- *School of Photogrammetry, Computer Vision and Laser Scanning.* International Centre For Mechanical Sciences (CISM), Udine, 14-16/4/2010.
- *Recent developments in image-based 3D reconstruction.* Tutorial at the VISMAC IAPR Summer schoool, 11 November 2010, Catania, IT.
- *Elements of Geometric Computer Vision* (Short course). Master VIBOT (Erasmus Mundus) at IUT Le Creusot - Université de Bourgogne, 8-29/10/2008.
- *Image-based rendering methods.* Summer school in Information Engineering, University of Padova, Brixen, July 2007.

- *Elements of Geometric Computer Vision.* Tutorial. 28th Annual Symposium of the German Association for Pattern Recognition, Berlin, September 2006.
- *Camera calibration: problems and algorithms.* Tutorial at the VISMAC IAPR Summer School, 29 September 2006, Palermo, IT.
- *Introduction to Computer Graphics.* Doctoral School in Information Engineering, University of Padova (I), July 2006.
- *Elements of Geometric Computer Vision.* University of California, Santa Barbara, Department of Electrical and Computer Engineering. Interactive Digital Multimedia IGERT Programme. June 2006.
- *Elements of Multiple View Geometry for Computer Vision.* Doctoral School in Information Engineering, University of Padova (I), June-July 2005.
- *Regularization Machines for Learning* Doctoral course "Modelli per l'Analisi e l'Elaborazione dei Dati", University of Verona, 2003.

PhD thesis supervised

- Abdul Salam Rasmi Asraf Ali. *CBCT Motion Correction by Optimization for Dental Applications*. Dottorato di ricerca in Ingegneria Industriale e dell'Informazione. Università degli Studi di Udine, XXXVI Ciclo.
- Eleonora Maset. *Advanced Methods for LiDAR and Photogrammetric Data Processing: from Procrustes Analysis to Deep Learning*. Dottorato di ricerca in Ingegneria Industriale e dell'Informazione. Università degli Studi di Udine, 2019.
- Federica Arrigoni. *Synchronization Problems in Computer Vision*. Dottorato di ricerca in Ingegneria Industriale e dell'Informazione. Università degli Studi di Udine, 2017.
- Luca Magri. *Stima di modelli ibridi per dati eterogenei*. Dottorato di Ricerca in Matematica e Statistica per le Scienze Computazionali. Università degli Studi di Milano, 2015.
- Francesco Malapelle. *Towards a Complete Pipeline for 3DS Conversion of Monocular Uncalibrated Images*. Dottorato di ricerca in Ingegneria Industriale e dell'Informazione. Università degli Studi di Udine, 2015.
- Valeria Garro. *Image localization and parsing using 3D structure*. Dottorato di Ricerca in Informatica. Dipartimento di Informatica, Università di Verona, 2013.
- Roberto Toldo. *Towards automatic acquisition of high-level 3D models from images*. Dottorato di Ricerca in Informatica. Dipartimento di Informatica, Università di Verona, 2013.
- Riccardo Gherardi. *Advances in 3D reconstruction*. Dottorato di Ricerca in Informatica. Dipartimento di Informatica, Università di Verona, 2010.
- Davide Moschini. *Tracking human motion with multiple cameras using articulated ICP with hard constraints*. Tesi di Dottorato di Ricerca. Dipartimento di Informatica, Università di Verona, 2009.
- Michela Farenzena. *Geometry and Appearance Modelling from Images*. Tesi di Dottorato di Ricerca. Dipartimento di Informatica, Università di Verona, 2007.
- Andrea Colombari. *Image-Based Techniques for View Synthesis and Video Editing*. Dottorato di Ricerca in Informatica. Dipartimento di Informatica, Università di Verona, 2006.
- Umberto Castellani. *Image-based modeling: from sensory data to 3D models*. Dottorato di Ricerca in Informatica. Dipartimento di Informatica, Università di Verona, 2003. Con V. Murino.

Master's and Laurea Thesis Supervision:

- Three Master's theses in Electronic Engineering and 53 theses in Computer Science (Master's, specialist and pre-reform degrees) supervised. Detailed list: `https://fusiello.github.io/index.php/Thesis_supervised.html`.

PROFESSIONAL SERVICE:

- **Scientific boards**
  - **Co-chair** of ISPRS WG II/1 (International Society for Photogrammetry and Remote Sensing), June 2016 - June 2022.
  - **Auditor** for the Italian Association for Computer Vision, Pattern Recognition and Machine Learning (CVPL), 2019-2020.
  - **Member** of the Executive Board of the italian chapter of Eurographics, from 2006 to 2009.
- **Editorial board**:
  - **Associate editor** of *IEEE Transactions on Image Processing*, since 2021.
  - **Scientific board member** of the "Informatica" book series, Franco Angeli, since 2011.
  - **Guest editor** (with Filippo Stanco and Domenico Tegolo), of the "Special Section: Selected Papers from the 19th International Conference on Image Analysis and Processing (ICIAP) 2017", IET Computer Vision, 2018.
  - **Associate editor** of *IET Computer Vision*, from 2013 to 2016.
  - **Associate editor** of *Pattern Recognition*, from 2006 to 2009.
  - **Guest editor** (with Hong Bin Zha, Hideo Saito e Vittorio Murino), of the "Special issue on 3-D Image Analysis and Modelling", IEEE Transactions on System, Man, and Cybernetics, August 2003.
- **Conference organization**:
  - **Workshops Chair** of the *European Conference on Computer Vision* (ECCV), August 23-28, 2020, Glasgow, UK (with A. Bartoli).
  - **Poster Chair** of the 40th *Annual Conference of the European Association for Computer Graphics*, May 6-10, 2019, Genoa, Italy (with O. Bimber).
  - **Organizing committee member**, *1st Workshop on Traditional Computer Vision in the Age of Deep Learning*, ICCV Workshops, October 16, 2021.
  - **Organizing committee member**, *Photogrammetric Computer Vision Workshop*, CVPR Workshops, June 16-17, 2019, Long Beach, USA.
  - **Organizing committee member**, *ISPRS Workshop: Photogrammetric 3D Reconstruction for Geo-Applications*, September 21-22, 2017, Wuhan, China.
  - **Area chair** of the 20th *International Conference on Image Analysis and Processing*, September 9-13, 2019, Trento, Italy (with A. Del Bue and F. Tombari).
  - **General chair** of the 6th *International Conference on 3D Vision*, 5-8 September 2018,Verona
  - **Area chair** of the 19th *International Conference on Image Analysis and Processing*, 11-15 September 2017, Catania, IT (with D. Fofi)
  - **Area chair** at the 18th *International Conference on Image Analysis and Processing*, 7-11 September 2015, Genova, IT (with M. Goesele)
  - **Workshop Chair** at the 12th *European Conference on Computer Vision*, 7-13 October 2012, Firenze, IT (with V. Murino)
  - **Poster Chair** at the 33rd *Annual Conference of the European Association for Computer Graphics (Eugraphics)*, 13-18 May 2012, Cagliari, IT (with M. Wimmer)
  - **General chair** of the 7th *Eurographics Italian Chapter Conference*, 22-23 October 2009, Verona, IT. (with U. Castellani, A. Giachetti and V. Murino).
  - **Chair** of the special session *Computer Vision for Augmented Reality and Computer Augmented Environments*, within the 13th International Conference on Image Analysis and Processing, 6-8 September 2005, Cagliari, IT (with V. Murino)
  - **Program chair** of the first "Convegno del Gruppo Italiano Ricercatori in Pattern Recognition" (GIRPR), 17-18 June 2002, Verona, IT.
  - **Chair** of the *Young Scientist Forum*, special session within the 10th *International Conference on Image Analysis and Processing*, 21-29 September 1999, Venezia, IT.

- **Program committee / Reviewer**:
  - *IEEE International Conference on Computer Vision and Pattern Recognition* (**CVPR**) since 2005 (no 2017).
  - *IEEE International Conference on Computer Vision* (**ICCV**) since 2005 (no 2017).
  - *European Conference on Computer Vision* (**ECCV**) since 2006.
  - *3D Vision Conference* (**3DV**) 2014-2017.
  - *International Conference on Image Analysis and Processing* (**ICIAP**) 2007, 2009, 2013, 2021.
  - *British Machine Vision Conference* (**BMVC**) 2006, 2007, 2017, 2022.
  - *International Conference on Image Processing* (**ICIP**) 2008, 2009, 2010.
  - *International Conference on Computer Vision/Computer Graphics Collaboration Techniques and Applications* (**MIRAGE**) 2007, 2009.
- **Keynotes and tutorials**:
  - Invited talk "The Long March of 3D Reconstruction: From Tabletop to Outer Space." *ECCV Workshop: 3D Reconstruction in the Wild*, September 14, 2018, Munich, Germany.
  - Invited talk "Automatic three-dimensional modelling from images." *Open Conference: At the Frontiers of Drawing. Experiences of Virtual Models and Immersive Spaces*, January 10, 2018, Gorizia, Italy.
  - Tutorial "Global Network Orientation by Synchronization." *ISPRS Technical Commission II Symposium*, June 3-7, 2018, Riva del Garda, Italy.
  - Invited talk "From structure from motion to videosynthesis." Annual Conference of the Italian Telecommunications and Information Theory Group (GTTI), June 23, 2009, Parma, Italy.
  - Invited talk "Detecting multiple structures in geometric data: the J-linkage approach." Study day "Big Data: a challenge rich in opportunities", June 5, 2013, University of Udine.
  - Keynote speech "Structure from motion: state of the art, examples and open issues." *ISPRS Technical Commission V Symposium*, 23-25 June 2014, Riva del Garda, IT.
  - Tutorial "Structure from motion: historical overview and recent trends." *18th International Conference on Image Analysis and Processing*, 7-11 September 2015, Genova, IT.
  - Tutorial "Elements of Geometric Computer Vision." *28th Annual Symposium of the German Association for Pattern Recognition*, 12 September 2006, Berlin, DE.
- **Other events**:
  - *Recent Developments in Digital Photogrammetry: 3D Acquisition and Automatic Information Extraction Techniques*. Advanced Professional Training, CISM, Udine, July 8, 2022 (coordinator with G. Guidi).
  - "*VISMAC international summer school*", Grado (GO) 13-17/6/2016 (organizing committee).
  - "*School of Photogrammetry, Computer Vision and Laser Scanning: paradigmi ed algoritmi.*" International Centre for Mechanical Sciences (CISM), Udine, 14-16/4/2010 (organizer with F. Crosilla)
  - Serie "*VIPS Advanced School on Computer Vision, Pattern Recognition and Image Processing*" at the Dipartimento di Informatica, Università di Verona, from 2004 to 2010 (organizer with V. Murino)
- **Journal Reviewer**:
  - Among others: *International Journal of Computer Vision, IEEE Transaction on Pattern Analysis and Machine Intelligence, IEEE Transaction on Circuits And Systems For Video Technology, IEEE Transactions on Multimedia, IEEE Transaction on System, Man, and Cybernetics, IEEE Journal on Oceanic Engineering, IEEE Transaction on Image Processing, IEEE Transaction on Medical Imaging, Machine Vision and Applications, Computer Vision and Image Understanding, Image and Vision Computing Journal, Pattern Recognition Letters, ISPRS Journal on Photogrammetry and Remote Sensing*
- **Doctoral Thesis Review and Examination**:
  - Giang Truong (Edith Cowan University, Australia, 2022)

  - Bruno Fanini (Sapienza University of Rome, 2019)
  - Adel Saleh Ali Alraimi (Universitat Rovira i Virgili, 2019)
  - Andrea Apicella (University of Naples Federico II, 2018)
  - Guido Averna (University of Palermo, 2018)
  - Francesco Michielin (University of Padua, 2014)
  - Marco Cipolla (University of Palermo, 2011)
  - Danda Pani Paudel (Université de Bourgogne, 2015)
  - Luca Cosmo (Università Ca' Foscari Venezia, 2015)
  - Dong Han (Politecnico di Milano, 2014),
  - Hoi Sim Wong (University of Adelaide, 2013)
  - Fabio Bellavia (Università di Palermo, 2011),
  - Andrea Albarelli (Università Ca' Foscari Venezia, 2010),
  - Thierry Molinier (Université de Bourgogne, 2009)
  - Luca Ballan (Università di Padova, 2009),
  - Stefano Arca (Università di Milano, 2007),
  - Simone Gasparini (Politecnico di Milano, 2007),
  - Pietro Zanuttigh (Università di Padova, 2007),
  - Elisabetta Del Ponte (Università di Genova, 2007),
  - Marco Zuliani (University of California Santa Barbara, 2006),
  - Miguel Arredondo (Heriot-Watt Univ. Edinburgh, 2004).
- **Research project reviewer**
  - Israel Science Foundation, 2021.
  - Université Bourgogne Franche-Comté, 2019.
  - Austrian Science Fund (FWF), 2010.
  - University of Applied Sciences and Arts of Southern Switzerland, 2009.
  - Futuro in Ricerca (FIRB), 2011.
  - Italian Committee for Research Evaluation (CIVR), 2005.
  - Research Projects of National Interest (PRIN), 2007.
  - Autonomous Province of Trento, 2007, 2009, 2013.

UNIVERSITY SERVICE:

- **Board member**, PhD programme in Industrial and Information Engineering, University of Udine, since 2013.
- **Chair** of the Engineering Admissions Examination Committee and TOLC coordinator, University of Udine, 2012-2014.
- **Board member**, PhD programme in Computer Science, University of Verona, 2000-2012.
- **Chair** of the Teaching Committee, Master's programme in Computer Science and Engineering, University of Verona, 2010-2011.
- **Member** of the Teaching Committee, Joint Teaching Committee and Self-Assessment Committee, Computer Science Degree Programme Board, University of Verona, 2006-2009.
- **Secretary** of the Computer Science Degree Programme Board, University of Verona, 2005.
- **Rector's Delegate** for the Language and Audiovisual Centre (CLAV), University of Udine, October 2014 - September 2019.
- **Director and initiator** of the Master program in Computer Game Development, University of Verona, 2009-2011.
- **Vice-President** of the Computer Science and Engineering Graduate Studies Board, University of Verona, 2010-2011.

- **Member** (elected) of the Faculty External Relations Committee, University of Verona, from 2006 to 2009.
- **Member** (elected) of the Graduate School of Sciences Engineering Medicine board, University of Verona, from 2006 to 2009.

October 21, 2026

Andrea Fusiello